Solve

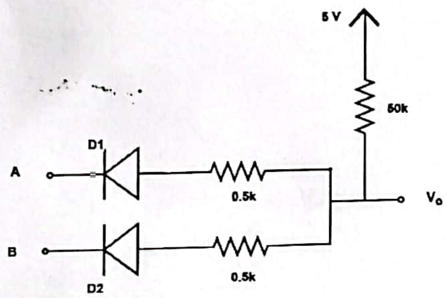
Assignment 1
Spring 25
CSE 350

Instruction:

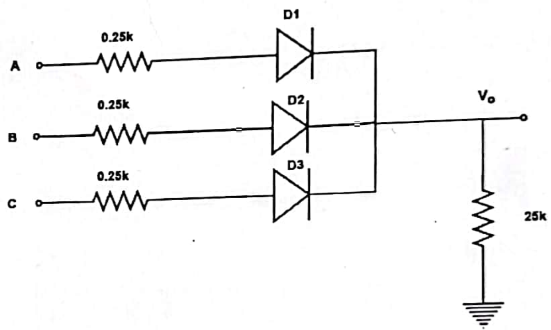
- ✓ Odd ID will solve the Set A and Even ID will solve Set B
- ✓ Solution must be in hand written format
- ✓ PDF file of the solution need to be submitted through google form

Set A

Answer the following questions from the given circuit. In this circuit input high logic and input low logic can be considered as 5 V and 0V. Diodes used in this circuit are Si diode. You can use constant voltage drop model (conduction voltage 0.7 V) for analysis of diodes.



Q1 (b)

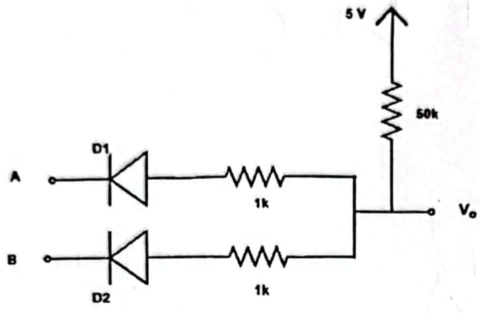


Q1 (c)

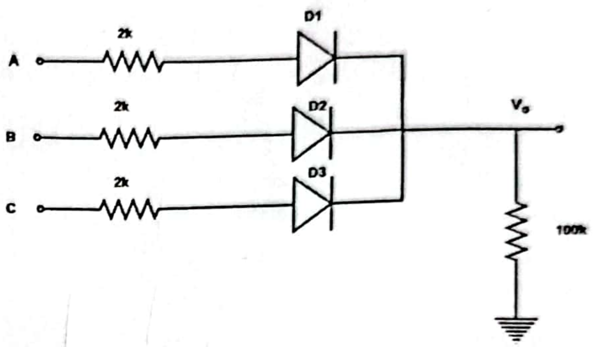
(a)	Implement the Boolean expression ($Y = A + BC$) using diode logic (DL) in terms of A, B and C.	[2]
(b)	Find the output voltage of the circuit (Fig: Q1(b)) for the input combination (A = 0, B= 1) and (A = 0, B = 0).	[4]
(c)	Which combination of inputs will show maximum power dissipation for the circuit (Fig: Q1(c))? Calculate the value of maximum power for this input combination.	[4]

Set B

Answer the following questions from the given circuit. In this circuit input high logic and input low logic can be considered as 5 V and 0V. Diodes used in this circuit are Si diode. You can use constant voltage drop model (conduction voltage 0.7 V) for analysis of diodes.



Q1 (b)

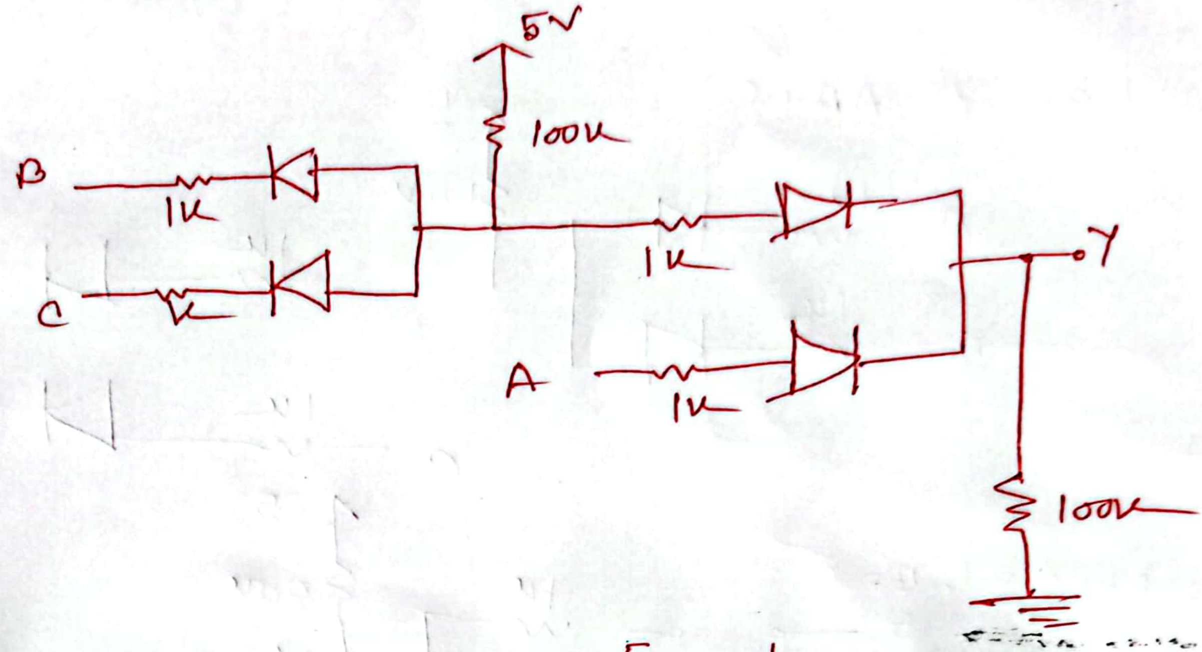


Q1 (c)

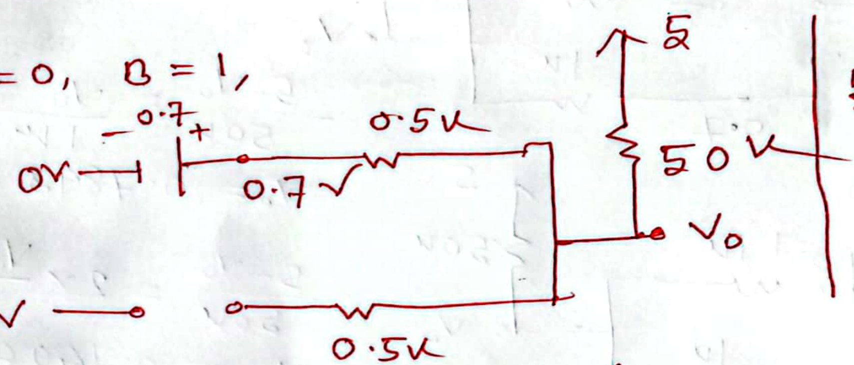
(a)	Implement the Boolean expression ($Y = AB + C$) using diode logic (DL) in terms of A, B and C.	[2]
(b)	Find the output voltage of the circuit (Fig: Q1(b)) for the input combination (A = 1, B= 0) and (A = 0, B = 0).	[4]
(c)	Which combination of inputs will show maximum power dissipation for the circuit (Fig: Q1(c))? Calculate the value of maximum power for this input combination.	[4]

Set A

1.a)



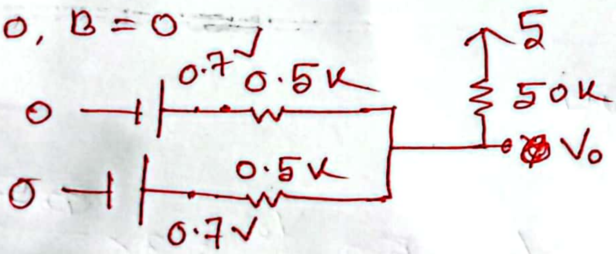
1.b) A = 0, B = 1,



$$\frac{5-V_o}{50k} = \frac{V_o-0.7}{0.5k}$$

$$V_o = 0.7425V$$

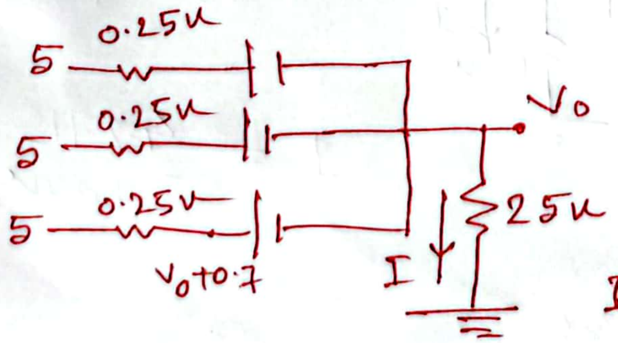
A = 0, B = 0



$$\frac{5-V_o}{50k} = \frac{V_o-0.7}{0.5k}\times 2$$

$$V_o = 0.7214V$$

1.c) (III) will show max power.



$$\frac{V_o}{25k} = 3\times\frac{5-V_o-0.7}{0.25k}$$

$$V_o = 4.2857V$$

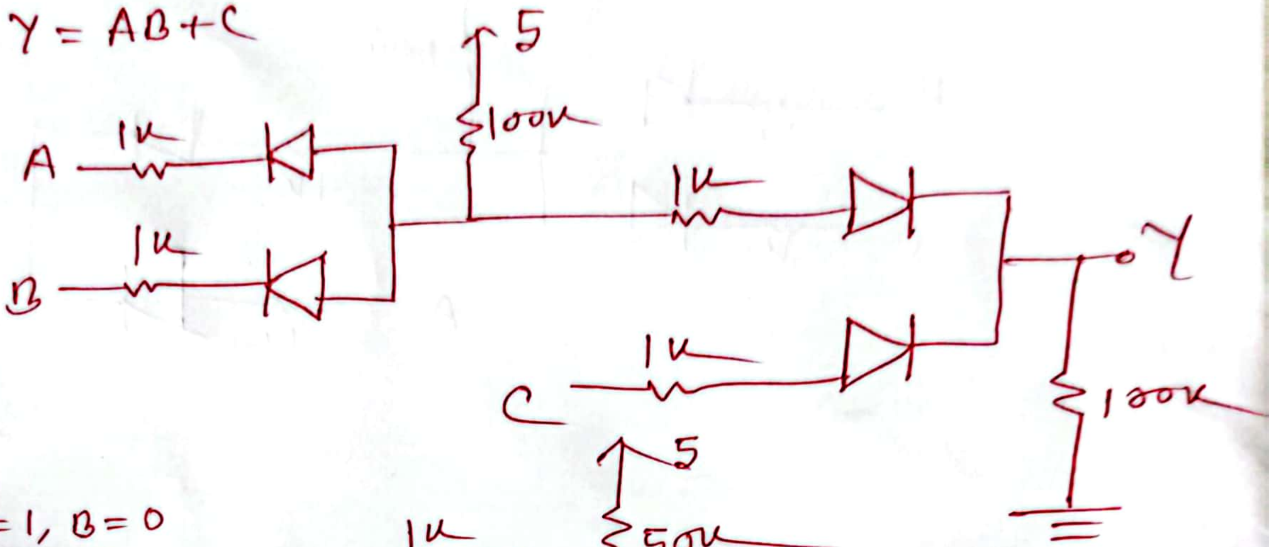
$$I = \frac{V_o}{25k}$$

$$P = (5-0)I$$

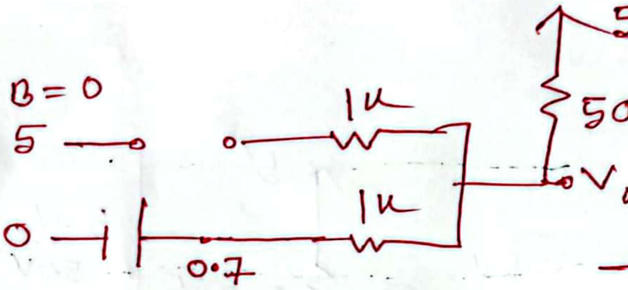
$$= (5-0)\times\frac{4.2857}{25k} = 0.8571\ mW$$

Set B

1.a) $Y = AB + C$



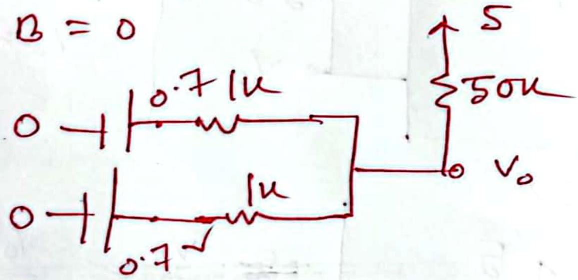
1.b) $A = 1, B = 0$



$$\frac{5 - V_0}{50k} = \frac{V_0 - 0.7}{1k}$$

$V_0 = 0.7843$ ✓

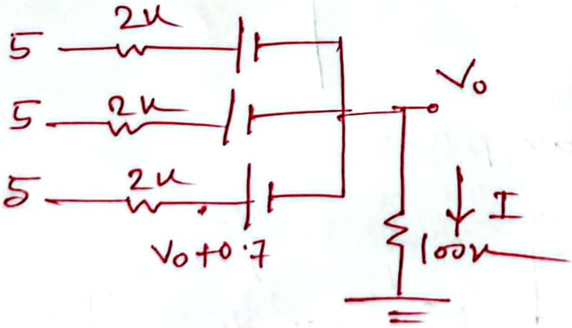
$A = 0, B = 0$



$$\frac{5 - V_0}{50k} = 2 \times \frac{V_0 - 0.7}{1k}$$

$V_0 = 0.7426$ ✓

1.c) (III) shows max power



$$\frac{V_0}{100k} = 3 \times \frac{5 - V_0 - 0.7}{2k}$$

$V_0 = 4.2715$ ✓

$$I = \frac{4.2715}{100k}$$

$P = (5 - 0) I$

$= 0.2135\ mW$